David Vayntrub

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Education

University of California, Riverside | B.S. in Computer Science
Sep 2024 - June 2027 (expected) | GPA: 3.84/4.00

Experience

StudySpot | Cofounding Engineer | May 2025 - Sep 2025

- Built an AI study frontend with Next.js and TypeScript, using Supabase and Anthropic to answer questions from course documents.
- Integrated the Canvas LMS API to import courses and documents, reducing onboarding time by an estimated 50% based on observed use.
- Improved document ingestion speed by 30% in timed testing by integrating LlamaParse; split UI redesign work into independent components with teammates.

ACM Riverside Forge | Software Engineer, club project | Jan 2025 - Mar 2025

- Developed speech recognition for a self-moving chess set on a 10-member ACM club team, using Python, SpeechRecognition and PyAudio to parse voice commands.
- Refined recognition of chess-specific phrases, reducing word error rate from 15% to 4%.

Projects

Open Source Poker Solver | Jul 2026 - Present

- Built a heads-up no-limit hold'em trainer combining a Rust CFR/neural solver with a Next.js interface, real-time EV feedback, adaptive drills and IndexedDB session persistence.
- Pinned model versions and artifact hashes per hand, verified response identity and stopped scoring when policies were unavailable; tested valid action distributions across 16,900 preflop rows and 5,997 postflop states.

PokerFly | Sep 2026 - Present

- Built a PyTorch poker-policy distillation experiment with FlyVis-constrained recurrent wiring and an interactive React, TypeScript and Three.js frontend backed by Node.js and Python.
- Evaluated policy imitation on 435 held-out decisions, reaching 85.5% agreement with the teacher's top action.

Bit Wizards | Cutie Hack | Nov 2025

- Built a Next.js and Blockly visual programming environment with 15+ custom syntax and control-flow blocks, JavaScript code generators and a sandboxed client-side runtime for immediate feedback.

Machine Learning Research Project | Aug 2022 - Mar 2024

- Processed 32,560+ census records with Python, pandas and NumPy, and compared classification methods in scikit-learn using five-fold cross-validation; authored a 20-page research paper.

Leadership

University of California, Riverside Chess Club | President | Sep 2026 - Present

- Grew first-week new membership by 100% year over year through marketing and equipment improvements; managed the club website.

Skills

Languages: Python, TypeScript, JavaScript, Rust, C++, Java, SQL, HTML/CSS
Frameworks and data: React, Next.js, Tailwind CSS, PyTorch, scikit-learn, pandas, NumPy, Supabase/PostgreSQL, IndexedDB
Tools and APIs: Git, REST APIs, Anthropic API, Gemini API, Canvas LMS API, LlamaParse